CCS6334
DIGITAL & COMPUTER FORENSICS

Lecture 10: Examining Browsers, E-mails and Messaging Systems

1

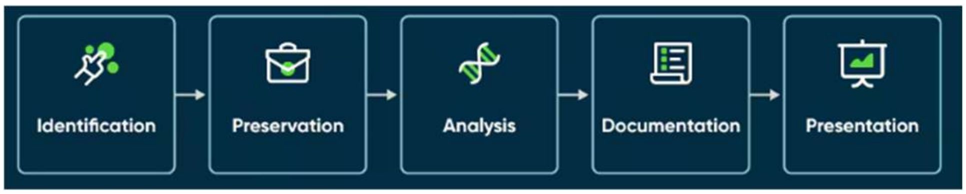
Learning Objectives

After studying this chapter, you should be able to:

- Locate and analyze digital evidence from Internet browsing activities and messaging systems for forensic investigation purposes.
- Conduct email analysis and efficiently process large email databases to extract relevant forensic evidence.

2

INTERNET BROWSING HISTORY



3

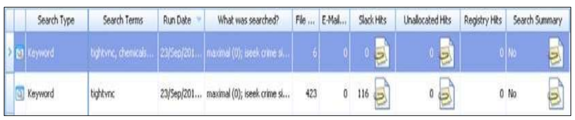
Typical Web-Browsing Behavior in Forensic Analysis

- Search Activities – Users look up people, events, organizations, accounts, or services.
- Locard's Principle – Every visit leaves digital traces exchanged between user and website.
- Website Logs – Servers (e.g., Hotmail, Yahoo) record IP addresses and login details.
- Email Evidence – Gmail stores messages in the cloud; POP clients and Windows caching enable local recovery.
- Cached Records – Browsers store HTML files, multimedia, and session data.
- History Databases – Track visited pages with timestamps.
- Cookies – Record websites visited and session details.
- Online Accounts & E-Commerce – Evidence of account access and banking activities.

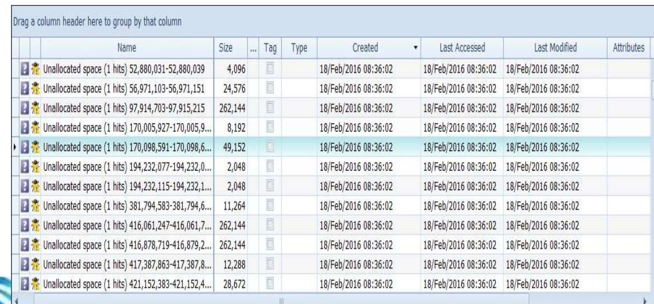
4

Recovering Browsing Artifacts from Slack & Unallocated Space

- Similar to analysis done on storage devices, browser artifacts can possibly be retrieved in slack space
- Some tools will automatically locate and index these items (e.g. ILook, Autopsy) to allow for further manual analysis by the forensic practitioner
- The main text references shows how ILook can inspect unallocated storage blocks and index them according to the type assumed (usually through header analysis)



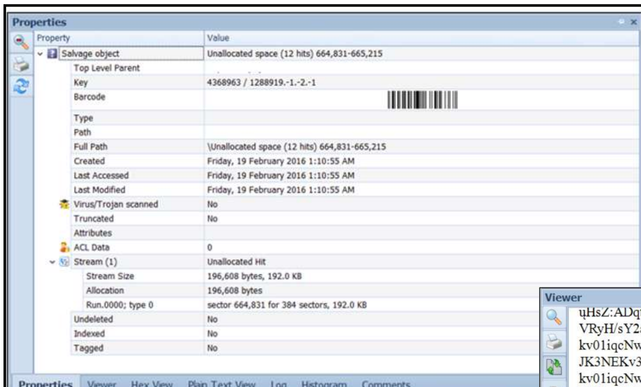
Search Type	Search Terms	Run Date	What was searched?	File ...	E-Mail...	Slack Hits	Unallocated Hits	Registry Hits	Search Summary
Keyword	tightvnc, chemicals...	23/Sep/201...	maximal (0); iseek crime si...	6	0	0	0	0	No
Keyword	tightvnc	23/Sep/201...	maximal (0); iseek crime si...	423	0	116	0	0	No



Drag a column header here to group by that column

Name	Size	...	Tag	Type	Created	Last Accessed	Last Modified	Attributes
Unallocated space (1 hits) 52,880,031-52,880,039	4,096				18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	
Unallocated space (1 hits) 56,971,103-56,971,151	24,576				18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	
Unallocated space (1 hits) 97,914,703-97,915,215	262,144				18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	
Unallocated space (1 hits) 170,005,927-170,005,9...	8,192				18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	
Unallocated space (1 hits) 170,098,591-170,098,6...	49,152				18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	
Unallocated space (1 hits) 194,232,077-194,232,0...	2,048				18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	
Unallocated space (1 hits) 194,232,115-194,232,1...	2,048				18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	
Unallocated space (1 hits) 381,794,583-381,794,6...	11,264				18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	
Unallocated space (1 hits) 416,061,247-416,061,7...	262,144				18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	
Unallocated space (1 hits) 416,878,719-416,879,2...	262,144				18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	
Unallocated space (1 hits) 417,387,863-417,387,8...	12,288				18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	
Unallocated space (1 hits) 421,152,383-421,152,4...	28,672				18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	18/Feb/2016 08:36:02	

5

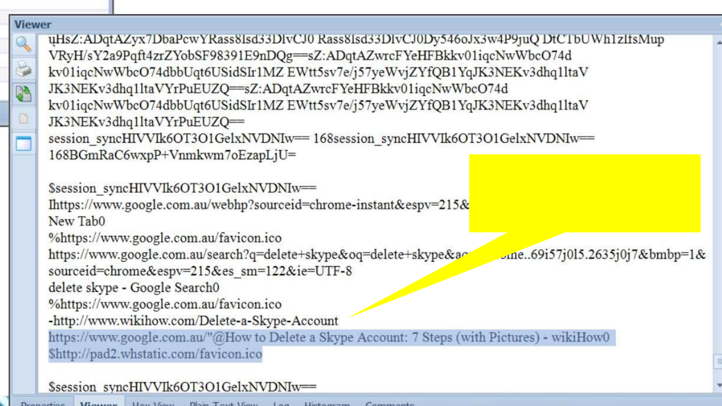


Properties

Property	Value
Salvage object	Unallocated space (12 hits) 664,831-665,215
Top Level Parent	
Key	4368963 / 1288919.-1.-2.-1
Barcode	
Type	
Path	
Full Path	\Unallocated space (12 hits) 664,831-665,215
Created	Friday, 19 February 2016 1:10:55 AM
Last Accessed	Friday, 19 February 2016 1:10:55 AM
Last Modified	Friday, 19 February 2016 1:10:55 AM
Virus/Trojan scanned	No
Truncated	No
Attributes	
ACL Data	0
Stream (1)	Unallocated Hit
Stream Size	196,608 bytes, 192.0 KB
Allocation	196,608 bytes
Run.0000; type 0	sector 664,831 for 384 sectors, 192.0 KB
Undeleted	No
Indexed	No
Tagged	No

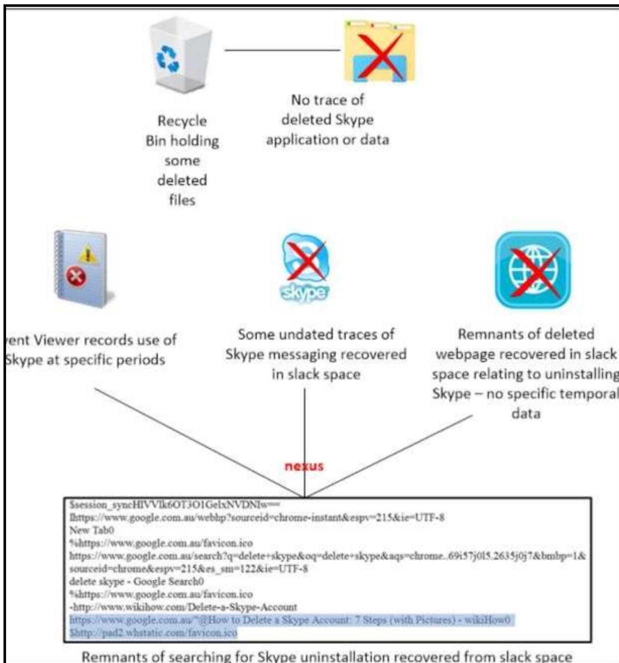
Properties of recovered Internet browsing from slack space

- In this example, data was recovered from slack file sectors using the term search terms
- The following screenshot highlights a website visited to delete a Skype account.
- This information was commensurate with the suspect's presumed attempt to delete the account to prevent future investigation of illegal activities involving communication with organized crime personalities:



A view of some of the readable data from Internet browsing records recovered from slack space

6

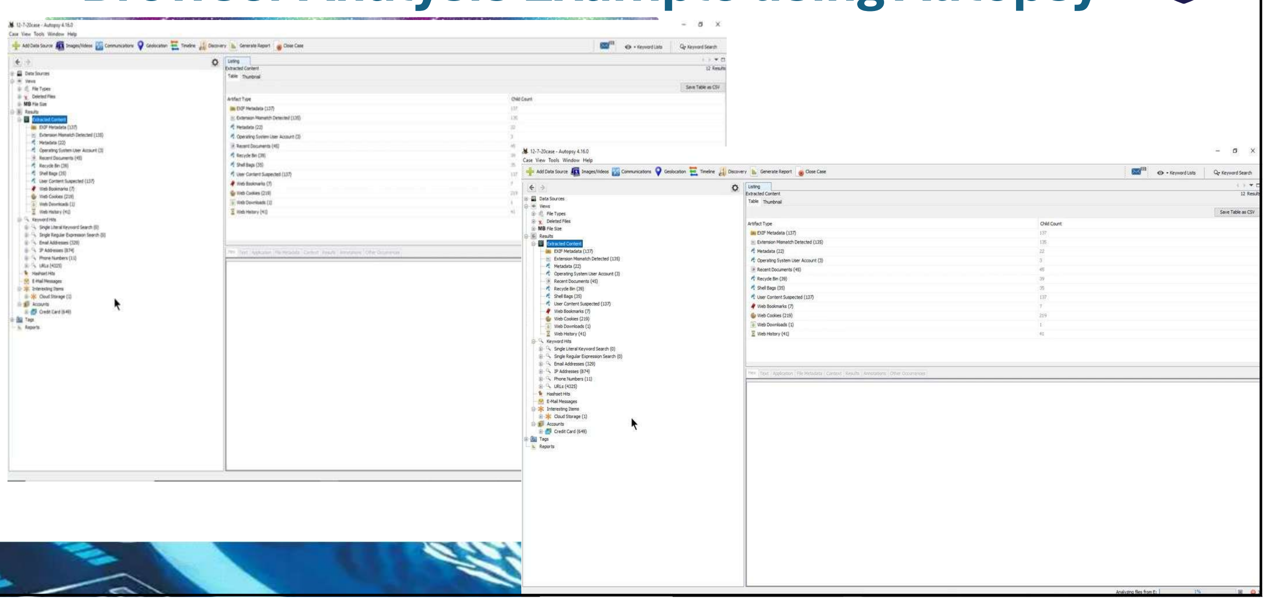


The nexus between data recovered from slack space to assist event reconstruction and an attempt to obfuscate the use of Skype

- Analysis of slack space shows history of a user trying to remove traces of a program (in this case Skype)
- Can we corroborate this information?
  - Check registry logs for HKCU installs to see installed/removed programs
  - Perform storage file carving to locate installation folder, executables, logs for Skype program
- Corroboration of these items might come in useful in reconstructing key events

7

Browser Analysis Example using Autopsy



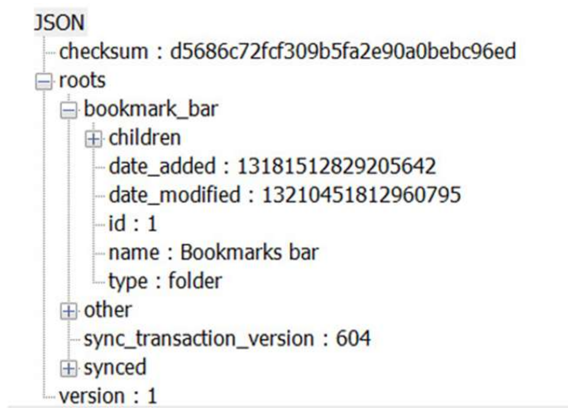
8

Browser Example: Google Chrome

- The first artifact we will look at is the **user's bookmarks**.
- The bookmarks allow the user to save web pages they find interesting and may give insight into the user's activity.
- You can find the bookmarks file at the following path:
- *%USERS%/AppData/Local/Google/Chrome/User Data/Default/Bookmarks*

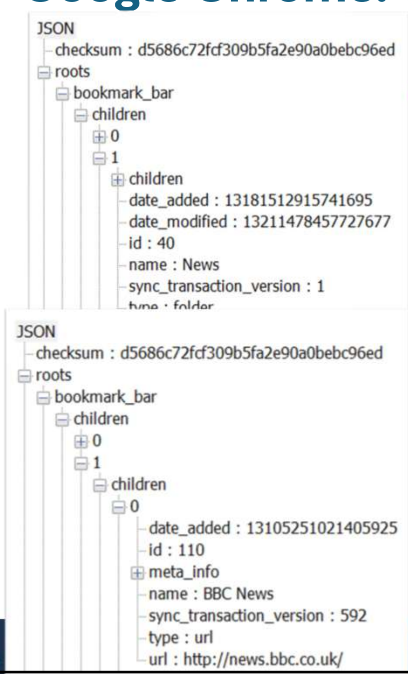
Here, we can see the following fields:

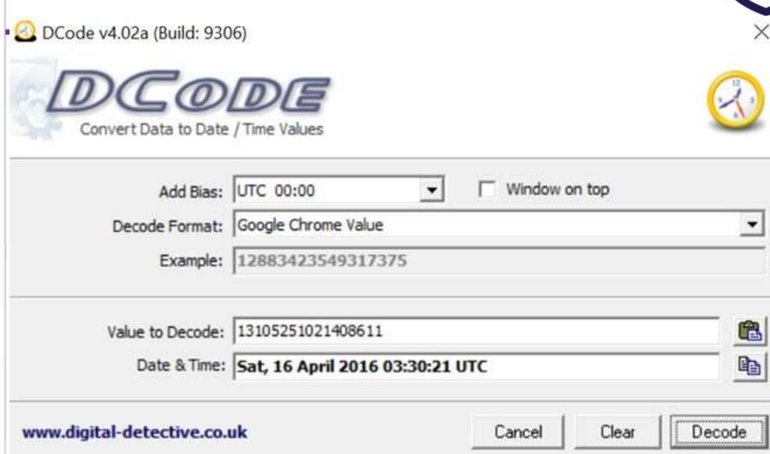
- Date added
- Last visited desktop
- The name of the bookmark
- The URL



9

Google Chrome: Understanding bookmarks





10

Understanding the Chrome history file

- The Google Chrome history file will be found at the following path:

%USERS%/AppData/Local/Google/Chrome/User Data/

- The history database contains quite a lot of information about the user's activity:
  - Downloads
  - Search terms that the user entered using the URL address bar.
  - Typed URLs will track the URLs the user typed into the address bar.
  - History

11

Google Chrome: Cookies

- A cookie is a file created by a website and stored on the user's system.
- Cookies are designed to track the user's activity, such as adding an item to a shopping cart or recording the pages the user has visited.
- The Google Chrome cookie file can be found at the following path:
  %USERS%/AppData/Local/Google/Chrome/User Data/Default

creation_utc	host_key	name	value	path	expires_utc	is_secure	is_httponly	last_access_utc	has_expires	is_pe
13211504229653934	.google.com	_ga		/gmail/about	13274576231000000	0	0	13211504229653934	1	1
13211504229654926	.google.com	_gid		/gmail/about	13211590631000000	0	0	13211504229654926	1	1
13211504361869670	.google.com	APISID		/	13274576361869670	0	0	13211568843104513	1	1
13211504373193089	mail.google.com	COMPASS		/mail/u/0	13212368374193089	1	1	13211504554421139		

Host Name	Path	Name	Value	Secure	HTTP Only	Last Access...	Created On	Expires
.google.com	/gmail/about	_ga		No	No	8/28/2019 22:17	8/28/2019 22:17	8/27/2021 22:17
mail-ads.google.com	/mail/u/0	COMPASS		Yes	Yes	8/28/2019 22:19	8/28/2019 22:19	9/7/2019 22:19
www.yahoo.com	/	flash_enabled		No	No	8/28/2019 22:21	8/28/2019 22:21	9/27/2019 22:21

12

Google Chrome: Cache & Passwords

- Chrome Cache View converts the data into a readable format.

gmail.html	https://www.google...	text/html	0	8/28/2019 15:17	8/28/2019 15:17			sffe	HTTP/1.1 302			private	172.217.14.100
s2	https://www.google...	text/javascript	14,685	8/28/2019 15:17	8/27/2019 14:47	8/27/2019 14:27	8/26/2020 14:47	sffe	HTTP/1.1 200	br	data_3 [253952]	public, max-age=31536000	172.217.14.100
about.html	https://mail.google....	text/html	0	8/28/2019 15:17	8/27/2019 21:40		8/28/2019 21:40	sffe	HTTP/1.1 301			public, max-age=86400	172.217.11.165
about.html	https://www.google...	text/html	0	8/28/2019 15:17	8/28/2019 04:21		8/29/2019 04:21	sffe	HTTP/1.1 301			public, max-age=86400	172.217.14.100
about.html	https://www.google...	text/html	15,504	8/28/2019 15:17	8/28/2019 15:17	7/19/2019 00:30	8/28/2019 15:17	sffe	HTTP/1.1 200	gzip	data_3 [303104]	private, max-age=3000	172.217.14.100

Passwords

- Passwords can be key to unlocking files or encryption.
- User's previously used passwords can be a treasure trove of information.
- Chrome has the option for a user to save passwords.
- You will find the password information in the Logon Data file, which can be found at the following path:
  %USERS%/AppData/Local/Google/Chrome/User Data/Default

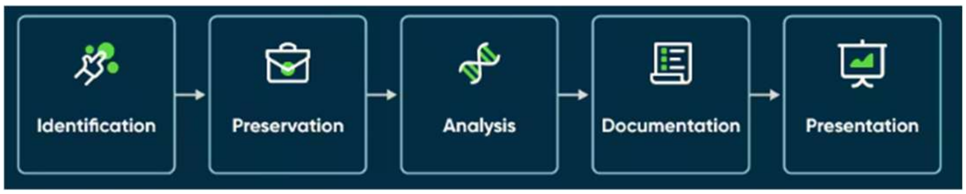
13

Challenges with Browser Forensics

- Incognito mode reduces availability of locally cached data
- Security/privacy focused browsers are becoming increasingly popular
- Variety of browsers available on different platforms
  - Chrome may be the most popular but aside from that is Brave, Opera, Mozilla Firefox, Microsoft Edge/IE, Mac Safari, Waterfox, Flashfox, Comodo IceDragon, Gnuzilla IceCat, Qihoo360, Baidu Browser, QQBrowser, etc.

14

MESSAGING SYSTEMS



15

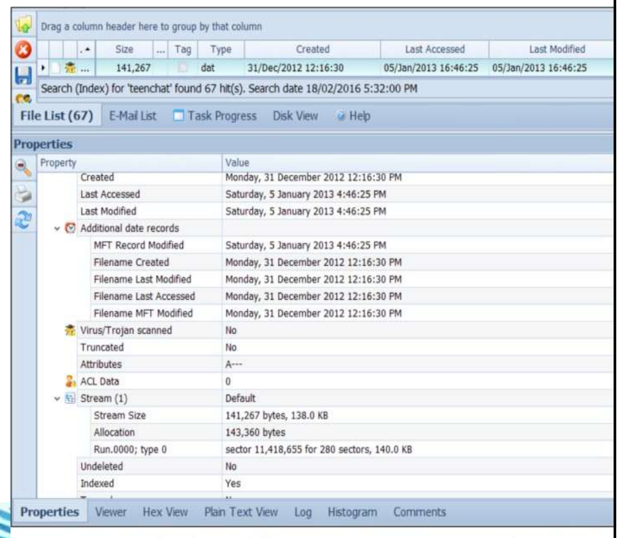
Messaging & Social Platforms as Digital Evidence

- Messaging apps and online platforms often contain valuable digital evidence on user activities, contacts, and intentions
- Widely used communication tools include WhatsApp, Telegram, Signal, Facebook Messenger, Instagram, WeChat, and web-based email services
- Chat rooms have resurged across computers, tablets, and mobile devices due to ease of use and convenience (e.g., Discord, Reddit, and Telegram groups)
- Social networking sites (Facebook, TikTok, Twitter (X), LinkedIn, and Snapchat) enable communication and community building
- Some chat platforms cater to all interests, including illegal or inappropriate activities, making them critical targets in digital investigations

16

Messaging & Social Platforms as Digital Evidence

- Browser based communications can be analysed via keyword searching or URL matching against browser history and logs
- The screenshot shows the file property sheet of recovered chat messages between a suspect and young persons through the social networking site → a site intended for teens and adolescents but often preyed upon by paedophiles and cyberstalkers



Property sheet of a record of an innocuous conversation using TeenChat

17

Messaging & Social Platforms as Digital Evidence

- Data relating to chat rooms is often logged and conversations and an exchange of multimedia files may also remain on the device, despite attempts by wrongdoers to delete and remove incriminating evidence
- Skype, a popular online mass communication program, by default, leaves a collection of spreadsheets that are helpful to the investigator

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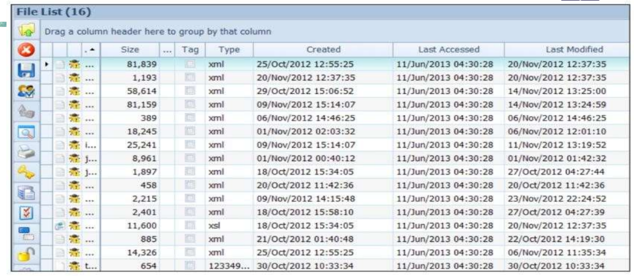
- Rival mass online meetup application, Zoom, also has many cached files in local storage
- Many of them stored according to OS platform *(\users\[name]\AppData\Roaming\Zoom* on Windows and *\Users\[name]\Library\Application Support\zoom.us\data* on Macs)
- A memory dump of their application also indicated weak spots in their application which resulted in the potential zero-day vulnerability on Windows platforms

18

Analysis Tools

- Certain forensic analysis tools will have features that target specific programs and can decipher their (local) storage and logs.
- Ilook has features that target Skype application logs

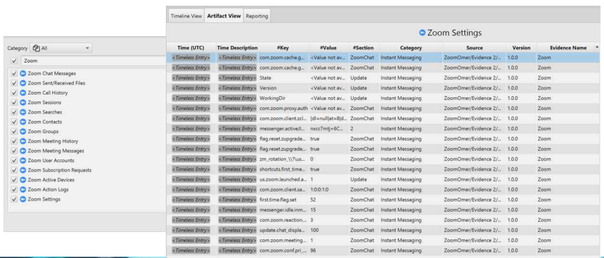


File List (16)

Drag a column header here to group by that column

.	Size	...	Tag	Type	Created	Last Accessed	Last Modified
...	81,839			xml	25/Oct/2012 12:55:25	11/Jun/2013 04:30:28	20/Nov/2012 12:37:35
...	1,193			xml	20/Nov/2012 12:37:35	11/Jun/2013 04:30:28	20/Nov/2012 12:37:35
...	58,614			xml	29/Oct/2012 15:06:52	11/Jun/2013 04:30:28	14/Nov/2012 13:25:00
...	81,159			xml	09/Nov/2012 15:14:07	11/Jun/2013 04:30:28	14/Nov/2012 13:24:59
...	389			xml	06/Nov/2012 14:46:25	11/Jun/2013 04:30:28	06/Nov/2012 14:46:25
...	18,245			xml	01/Nov/2012 02:03:32	11/Jun/2013 04:30:28	06/Nov/2012 12:01:10
i...	25,241			xml	09/Nov/2012 15:14:07	11/Jun/2013 04:30:28	11/Nov/2012 13:19:52
j...	8,961			xml	01/Nov/2012 00:40:12	11/Jun/2013 04:30:28	01/Nov/2012 01:42:32
j...	1,897			xml	18/Oct/2012 15:34:05	11/Jun/2013 04:30:28	27/Oct/2012 04:27:44
...	458			xml	20/Oct/2012 11:42:36	11/Jun/2013 04:30:28	20/Oct/2012 11:42:36
...	2,215			xml	09/Nov/2012 14:15:48	11/Jun/2013 04:30:28	23/Nov/2012 22:24:52
...	2,401			xml	18/Oct/2012 15:58:10	11/Jun/2013 04:30:28	27/Oct/2012 04:27:39
...	11,600			xsl	18/Oct/2012 15:34:05	11/Jun/2013 04:30:28	20/Nov/2012 12:37:35
...	885			xml	21/Oct/2012 01:40:48	11/Jun/2013 04:30:28	22/Oct/2012 14:19:30
...	14,326			xml	25/Oct/2012 12:55:25	11/Jun/2013 04:30:28	06/Nov/2012 11:35:34
t...	654			123349...	30/Oct/2012 10:33:34	11/Jun/2013 04:30:28	30/Oct/2012 10:33:34

Recovered files relating to Skype activity



- Artifast has a program that can read and decipher Zoom data stores showing last Zoom call details and messages passed in the session
- Axiom Magnet also has similar functionality
- MSAB XRY / XAMN – Mobile chat recovery, timeline reconstruction, cloud artefacts

19

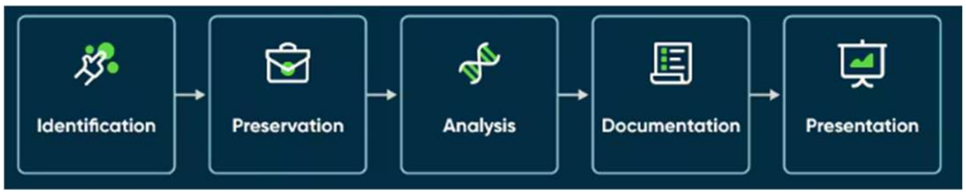
What About Cloud-based Messaging Systems?

- Some messaging systems work without having a standalone application (e.g. Google Meet, Google Hangouts, etc.)
- These systems can either rely on the analysis of browser history to obtain evidence OR employ specialized tools for cloud analysis, e.g. ESICloud Forensics for Google platform
- NOTE: some web-based messaging systems do have mobile applications and, the mobile application stores are analysed instead (e.g. FB Messenger, Whatsapp, etc)
- Other tools:
  - Cellebrite Inspector (Cloud) – Cloud-based messaging data (Google, iCloud, Meta)
  - Oxygen Cloud Extractor – Cloud chat backups and metadata
  - Magnet AXIOM Cloud – WhatsApp, Facebook Messenger, Instagram DM, Telegram

Google Meet

20

EMAIL SYSTEMS FORENSICS



21

Understanding email protocols

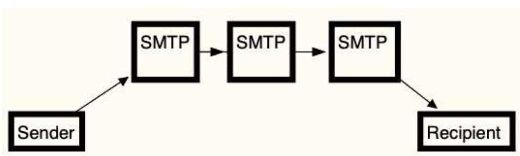
- An email protocol is a standard used to allow two computer hosts to exchange email communication.
- When an email is sent, it travels from the sender's host to an email server.
- The email server can forward the email through a series of relays until it arrives at an email server close to the recipient's host.
- The recipient will receive a notification stating that an email is available; the recipient will then reach out to the email server to get the email.

22

Understanding email protocols

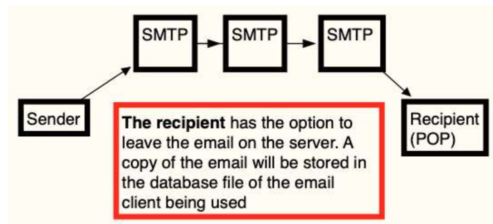
SMTP – Simple Mail Transfer Protocol

- SMTP is the protocol for email transmission. It is an internet standard based on RFC 821 but was later updated to RFC 3207, RFC 5321/5322



Post Office Protocol

- POP3 is the standardized protocol that allows users to access their inbox and download emails.
- POP3 is specifically designed only to receive emails; the system does not allow users to send emails.

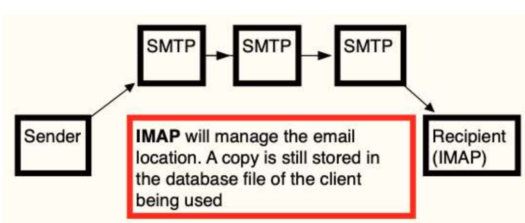


23

Understanding email protocols

IMAP – Internet Message Access Protocol

- IMAP is the Internet Message Access Protocol and is a standard protocol used by an email client to access emails on an email server.
- The protocol was designed to allow complete inbox management with multiple clients



Web-based email

- Web-based email is a service the user accesses with a web browser. Standard webmail providers are Gmail, Yahoo Mail, and Outlook/Hotmail.
- Some internet service providers also provide an email account that the user can access with a web browser

24

Decoding email (Understanding the email message format)

- An email has many unique identifiers for a digital forensic investigator to identify and track down.
- The mailbox and domain name, along with the message ID, will allow a digital forensic investigator to serve judicially approved subpoenas/search warrants on the vendor to follow any investigative leads.
- The vast majority of email users are only familiar with basic email information, such as this:

Subject:	background checks
Date:	07/19/2025 23:39:57 +0
Sender:	alison@m57.biz
Recipients:	jean@m57.biz

25

Decoding email (Understanding the email message format)

-----HEADERS-----

Return-Path: simsong@xy.dreamhostps.com

X-Original-To: jean@m57.biz

Delivered-To: x2789967@spunkymail-mx8.g.dreamhost.com

Received: from smarty.dreamhost.com (sd-green-bigip-81.dreamhost.com [208.97.132.81]) by spunkymail-mx8.g.dreamhost.com (Postfix) with ESMTP id E32634D80F for <jean@m57.biz>; Sat, 19 Jul 2025 16:39:57 -0700 (PDT)

Received: from xy.dreamhostps.com (apache2-xy.xy.dreamhostps.com [208.97.188.9]) by smarty.dreamhost.com (Postfix) with ESMTP id 6E408EE23D for <jean@m57.biz>; Sat, 19 Jul 2025 16:39:57 -0700 (PDT)

Received: by xy.dreamhostps.com (Postfix, from userid 558838) id 64C683B1DAE; Sat, 19 Jul 2025 16:39:57 -0700 (PDT)

To: jean@m57.biz From: alison@m57.biz

subject: background checks

Message-Id: 20250719233957.64C683B1DAE@xy.dreamhostps.com

Date: Sat, 19 Jul 2025 16:39:57 -0700 (PDT)

26

Decoding email (Understanding the email message format)

- The email header shows where the email originated from and what servers it touched upon. Starting from the bottom, we can see the Message-Id field:
- Message-Id: <20250719233957.64C683B1DAE@xy.dreamhostps.com>

The following two Received lines identify the subsequent servers on the path to the destination:

Received: from smarty.dreamhost.com (sd-green-bigip-81.dreamhost.com [208.97.132.81])

by spunkymail-mx8.g.dreamhost.com (Postfix) with ESMTP id E32634D80F for <jean@m57.biz>;

Sat, 19 Jul 2025 16:39:57 -0700 (PDT)Received: from xy.dreamhostps.com (apache2-xy.xy.dreamhostps.com [208.97.188.9])

by smarty.dreamhost.com (Postfix) with ESMTP id 6E408EE23D for <jean@m57.biz>;

Sat, 19 Jul 2025 16:39:57 -0700 (PDT)

27

Decoding email (Understanding the email message format)

- There are optional fields that you may come across in your investigations.
- These fields typically start with an X–

X-Priority: 3
X-Mailer: PHPMailer 5.2.9 (https://github.com/PHPMailer/PHPMailer/)
Message-Id: ff176aaf06e2f6958ada6e2d3c43b095@x3.netcomlearning.com
X-Report-Abuse: Please forward a copy of this message, including all headers, to abuse@mandrill.com
X-Report-Abuse: You can also report abuse here: http://mandrillapp.com/contact/abuse?id=30514476.1925a088d66f450cb25a4034f3ec6942 X-Mandrill-User: md_30514476

28

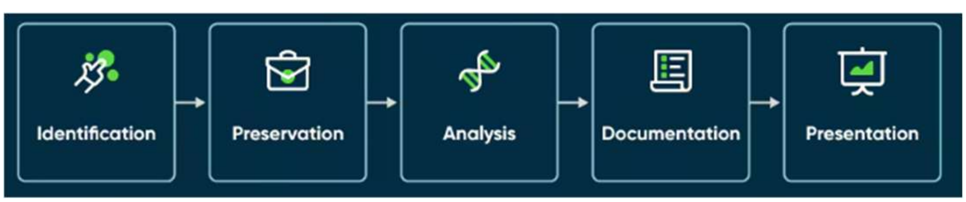
Email attachments

- MIME is the acronym for Multipurpose Internet Mail Extensions
- Internet standard for allowing emails to accept text other than ASCII, binary attachments, multi-part message bodies, and non-ASCII base header information.
- MIME indicated with the following:

MIME-Version: 1.0

29

CLIENT-BASED & WEBMAIL EMAIL ANALYSIS



30

Understanding client-based email analysis

- A user has access to many email clients to retrieve, read, and send emails.
- Depending on whether you're in the consumer or commercial environment, you may encounter different email clients.
- In the consumer market, you will find that Microsoft Outlook/Outlook Express will prevail because it is preinstalled on the system

31

Exploring Microsoft Outlook/Outlook Express

- Previous version of Outlook stores email information in several file types, such as pst, .mdb, and .ost. We will find the PST file on the user's hard disk at the following path:

\Users\$USER$\AppData\Local\Microsoft\Outlook

Outlook (New) for Windows 11:

C:\Users\<username>\AppData\Local\Packages\Microsoft.OutlookForWindows_*\LocalCache

- Cloud-centric, limited local artefacts
- No traditional PST/OST files
- Evidence mainly resides in cloud accounts (Microsoft 365, Outlook.com)
- 📌 *Forensics impact:* Requires cloud acquisition (e.g., Microsoft 365 audit logs, mailbox exports).

32

Mozilla Thunderbird

- Thunderbird is a free, open-source email client provided by Mozilla.
- Thunderbird will store emails within a .MBOX file.
- The MBOX format is a generic term for a family of file formats used to store emails. It will keep all the emails from folder into a singular database file. By default, the examiner can find the MBOX file in the following path:
- $USERNAME$\AppData\Roaming\Thunderbird\Profiles

```
u2xziaos.default-release (106)
├─minidumps (0)
├─crashes (1)
│ └─events (0)
├─extensions (1)
├─calendar-data (4)
├─storage (12)
│ └─permanent (12)
│   └─chrome (12)
│     └─idb (11)
│       └─3870112724rsegmnoittet-es.files (0)
├─ImapMail (16)
│ └─imap.mail.yahoo.com (15)
└─Mail (4)
  └─Local Folders (4)
```

Name	Type
.. = imap.mail.yahoo.com (15)	
. = INBOX (3)	mbox
Banks.eml	eml
New sign in on thunderbird.eml	eml
Re: midgets.eml	eml

33

Mozilla Thunderbird

- When we look in the folder, we will see the following files:

Archive.msf Archives.msf Bulk Mail.msf Draft.msf Drafts.msf INBOX	INBOX.msf msgFilterRules.dat Sent-1.msf Sent.msf Templates.msf Trash.msf

34

Understanding WebMail analysis

- WebMail is just another internet artifact for conducting browser analysis.
- Suppose the digital forensic investigator wants to investigate the user's use of web-based email.
- In that case, they will have to analyze the temporary internet files or the internet cache on the user's system.
- The temporary internet files/cache contains images, text, or any web page component the user has viewed in their browser.
- Before look into the cache, look into the internet history of the installed browser to see if the user has accessed web-based email.
- For the Chrome browser, you will find the history stored in a SQLite database named History at the following path:

$USER$\AppData\Local\Google\Chrome\User Data\Default

35

Understanding WebMail analysis

19	08/28/2019 22:19:39 +0	Inbox (2) - badguyneedslove@gmail.com - Gmail	https://mail.google.com/mail/?pc=topnav-about-n-en

The analysis of the **History** database shows the user accessed the Gmail web-based service

Found this in the internet cache for the Google Chrome browser, which the examiner can find at the following location:

$USER$\AppData\Local\Google\Chrome\User Data\Default\History Provider Cache

36

Understanding WebMail analysis

- If we keep searching for the email address we found in the cache, badguyneedslove@gmail.com, we may find other artifacts, such as the following:

{"endpoint_info_list":[{"endpoint":"**smtp:badguy27@yahoo.com**",
"c_id":"d24c.2d00",
"c_name":"**Joe Badguy Smith**"},
{"endpoint":"**smtp:badguyneedslove@gmail.com**",
"c_id":"e80f.5b71","c_name":"John Badguy Smith"},
{"endpoint":"**smtp:yahoo@mail.comms.yahoo.net**",
"c_id":"624f.10f0","c_name":"Yahoo! Inc."}]}

37

Understanding WebMail analysis

```
Mozilla (1,505)
└─Firefox (1,505)
   └─Profiles (1,504)
      ├─55abhq00.default-release (1,504)
      │  ├─safebrowsing (50)
      │  │  └─google4 (10)
      │  ├─jumpListCache (5)
      │  ├─startupCache (236)
      │  ├─cache2 (1,162)
      │  │  ├─entries (1,160)
      │  │  └─doomed (0)
      │  ├─thumbnails (0)
      │  ├─OfflineCache (1)
      │  └─safebrowsing-updating (49)
      │     └─google4 (9)
      └─cqr6ioib.default (0)
```

```
"matches": [
  {
    "lookupId": "badguyneedslove@gmail.com",
    "personId": [
      "114987255021342983529"
    ]
  }
],
"people": {
  "114987255021342983529": {
    "personId": "114987255021342983529",
    "metadata": {
      "lastUpdateTimeMicros": "1567030765000",
      "identityInfo": {
        "originalLookupToken": [
          "badguyneedslove@gmail.com"
        ],
        "sourceIds": [
          {
            "container": "PROFILE",
            "id": "114987255021342983529",
            "sourceEtag": "#3koZ3UbbsLY=",
            "containerType": "PROFILE"
```

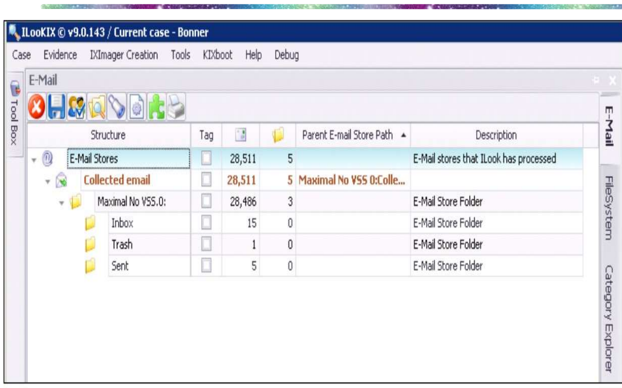
38

E-mail Analysis Tools

- E-mail evidence may originate from local computers or large network servers, often containing vast numbers of messages and attachments that require specialised tools for efficient selection, management, and analysis.
- E-mail store files include formats such as **.PST, .OST, .EML, .MSG, .NSF, .MBOX, .MBS**, and HTML-based e-mails, which may contain single messages or complete folder structures.
- Forensic tools (e.g., ILookIX) deconstruct all e-mail store types so they can be uniformly displayed and analysed within an E-mail Explorer, regardless of client or file format.
- A homogeneous e-mail object model based on Internet e-mail RFC standards standardises analysis across different clients (e.g., Outlook, Lotus Notes), enabling efficient searching, linking, and interpretation of e-mail evidence

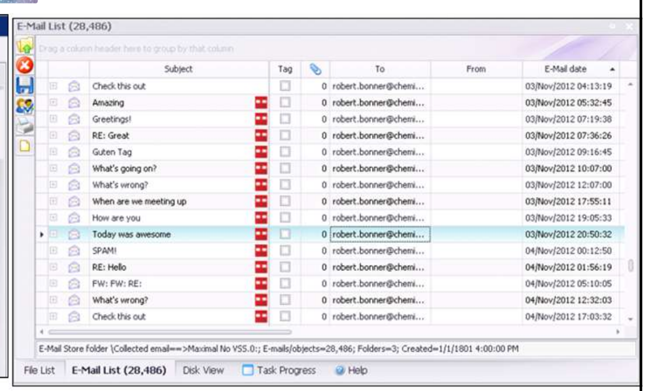
39

E-mail Analysis Tools (Cont.)



E-mail directory structure viewed in the ILookIX e-mail viewer

ILookIX deconstructs and organises over 28,000 e-mail messages and attachments from a single account into a structured view that mirrors the original device's e-mail directory, files, and attachments.



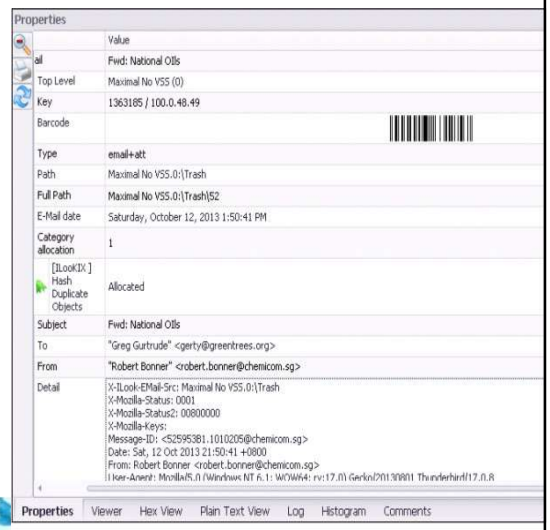
Viewing e-mails by subject

- Practitioners can view e-mail messages, contacts, attachments, and related items, with messages displayed in the List Pane when an e-mail store or its subfolder is selected

40

E-mail Analysis Tools (Cont.)

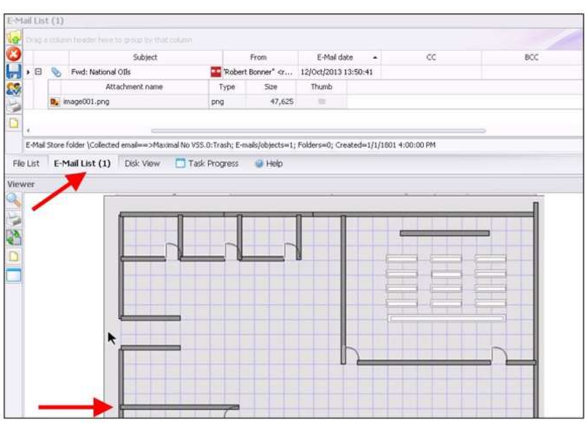
- E-mail file property sheets provide important file and metadata information about messages and are invaluable exports for case preparation
- These property sheets are basically automated extraction of header fields from the email stores
- Tools like ILook automatically categorize and index emails such that they can be searched using the functions provided (useful for bulk analysis

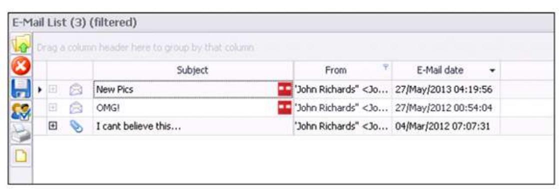


E-mail property sheet

41

E-mail Analysis Tools (Cont.)


Viewing e-mail attachments



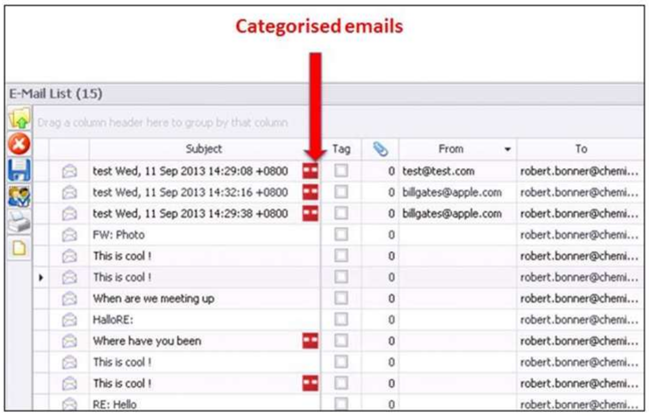
Headers, attachments, and e-mail body

Cataloging e-mail evidence of potential relevance

- By selecting the **E-Mail List**, the practitioner has more options for viewing attachments in the context of the messages to which they were originally attached

42

E-mail Analysis Tools (Cont.)



Categorizing e-mails of potential relevance

- E-mail messages can be included in the same categories as files or in separate categories specifically created for correspondence, as determined by the practitioner.
- Whole e-mail stores or subfolders therein can be added to categories by selecting the store or folder in the e-mail explorer and selecting Save Messages to My Categories

Good housekeeping seen by cataloging evidence of value during the selection process

43

Summary

What was covered today:

- Key processes for locating and recovering digital evidence relating to records of personal communications, including e-mails and browsing records stored on computer devices

44

45